Thermodynamics Tutorial 2

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Question 1

1. Determine the specific kinetic energy of a mass whose velocity is 45 m/s, in kJ/kg

Question 2:

A car is accelerated from rest to 85 km/h in 5 s. Would the energy transferred to the car be different if it were accelerated to the same speed in 3.5 s?

Question 3:

A 5 kg system was taken from 50°C to 150°C. Given the specific heat capacity of the system is Cp = 6000 J/kg, determine;

a) How much energy in the form of heat was added to the system to produce this temperature increase?
b) If the system received the heat over a period of 10 seconds, calculate the heat transfer rate in kW.

Question 4:

A steel rod of 0.8 cm diameter and 21 m length is stretched 3 cm. Young's modulus for this steel is 21 kN/cm^2. How much work, in kJ, is required to stretch this rod?

Question 5:

Determine the torque applied to the shaft of a car that transmits 250 hp (1 hp$\approx$ 745.7 W) and rotates at a rate of 3200 rpm.

Question 6:

A 100 kW engine and an 80 MW power plant are each used to lift a 500,000 kg mass 100 meters vertically. Calculate the minimum time (in seconds) it would take for each power source to complete this task.

Question 7:

A vertical piston-cylinder device contains water and is being heated. During the process, 70 kJ of heat is transferred to the water, and heat losses from the side walls amount to 9 kJ. The piston rises as a result of evaporation, and 6 kJ of work is done by the vapor. Determine the change in the energy of the water for this process.

Question 8

Consider a room that is initially at the outdoor temperature of 30°C. The room contains a 50-W lightbulb, a 120-W TV set, a 300-W refrigerator, and a 1200-W iron. Assuming no heat transfer through the walls, determine the rate of increase of the energy content of the room when all of these electric devices are on.

Question 9:

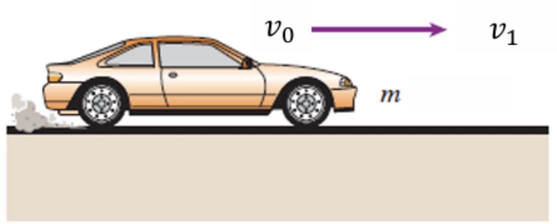
Consider a 2500-kg car cruising at constant speed of 80 km/h. Now the car starts to pass another car by accelerating to 110 km/h in 5 s. (a) Determine the additional power needed to achieve this acceleration. (b) What would your answer be if the total mass of the car were only 800 kg?

Analysis:

The work needed to accelerate a body is simply the change in the kinetic energy of the body.

$$W_a = \Delta E_k = \frac{1}{2}m(\Delta v^2) = \frac{1}{2}m(v_1^2 - v_0^2)$$

and $\dot{W}_a = \frac{W_a}{\Delta t}$



At $t_0, v_0 = 80\ \text{km/h}$

At $t_1, v_1 = 110\ \text{km/h}$

Question 10:

Consider a river flowing toward a lake at an average velocity of 4 m/s at a rate of 600 m^3/s at a location 80 m above the lake surface. Determine the total mechanical energy of the river water per unit mass and the power generation potential of the entire river at that location.